

Solution 03
Linear Quadratic Regulator

Exercise 1

Re-engineer the parameter $q \in \mathbb{R}$ of the stage cost function

$$\ell(x, u; q) = qx^2 + u^2 \quad (1)$$

such that the solution of the infinite horizon optimal control problem

$$\min_{x_i, u_i} \sum_{i=0}^{\infty} \ell(x_i, u_i) \quad (2a)$$

$$\text{s.t. } x_{i+1} = x_i + u_i, \quad i \in \mathbb{N}^+ \quad (2b)$$

$$x_0 = x(k) \quad (2c)$$

is given by the control law $u(k) = -\frac{1}{3}x(k)$.

Solution:

From Infinite Horizon Control (Lec 2, Slide 40), we have $u(k) = F_{\infty}x(k)$. It means that we want $F_{\infty} = -\frac{1}{3}$. We have

$$F_{\infty} = -(B^T P_{\infty} B + R)^{-1} B^T P_{\infty} A,$$

where we know everything apart from P_{∞} . Indeed, from system dynamics, we have that $A = 1$, $B = 1$ and from the cost function $R = 1$. Therefore, we can find the value of P_{∞} by solving

$$\begin{aligned} F_{\infty} &= -\frac{P_{\infty}}{P_{\infty} + 1} \stackrel{!}{=} -\frac{1}{3}, \\ \rightarrow P_{\infty} &= \frac{1}{2}. \end{aligned}$$

We can insert that value inside the Algebraic Riccati Equation (ARE) (Lec. 2, Slide 41)

$$P_{\infty} = A^T P_{\infty} A + Q - A^T P_{\infty} B (B^T P_{\infty} B + R)^{-1} B^T P_{\infty} A,$$

and we get

$$\begin{aligned} P_{\infty} &= P_{\infty} + q - \frac{P_{\infty}^2}{P_{\infty} + R}, \\ \rightarrow q &= \frac{P_{\infty}^2}{P_{\infty} + R} = \frac{1}{6}. \end{aligned}$$

It means that if the stage cost is defined as $\ell(x, u) = \frac{1}{6}x^2 + u^2$, then the optimal control input for the infinite horizon optimal control problem will be $u(k) = -\frac{1}{3}x(k)$.

Exercise 2

Use linear quadratic regulator (LQR) design in order to approximately minimize the infinite horizon cost

$$J(x_0) = \sum_{k=0}^{\infty} x(k)^2 + \frac{3}{4} u(k)^2 \quad (3)$$

with respect to the nonlinear system

$$x(k+1) = x(k) + x(k)^2 + u(k), \quad x(0) = x_0, \quad (4)$$

and to investigate the resulting closed-loop stability properties.

1. Linearize (4) around the equilibrium point $x_s = 0$ and $u_s = 0$.
2. Design an LQR controller $u(k) = F_{\infty} x(k)$ based on the linearized system that approximately minimizes the infinite horizon cost (3).
3. Which stability properties of the origin $x_s = 0$ for the resulting linear closed-loop system can be directly concluded?
4. Which stability properties of the origin $x_s = 0$ for the resulting nonlinear closed-loop system can be concluded?

Solution:

1. To linearize the system, we have to calculate the Jacobians. We have

$$A = \left. \frac{\partial f(x, u)}{\partial x} \right|_{x_s=0, u_s=0} = 1 \text{ and } B = \left. \frac{\partial f(x, u)}{\partial u} \right|_{x_s=0, u_s=0} = 1, \quad (5)$$

where we defined $f(x(k), u(k)) := x(k) + x(k)^2 + u(k)$. The linear system can be written as

$$x(k+1) = x(k) + u(k). \quad (6)$$

2. We calculate the LQR controller for the discrete-time linear system described by

$$x(k+1) = Ax(k) + Bu(k) \quad (7)$$

to minimize the cost function defined as

$$\sum_{k=0}^{\infty} x(k)^T Q x(k) + u(k)^T R u(k). \quad (8)$$

The LQR controller $u(k) = F_{\infty} x(k)$ is defined by

$$F_{\infty} = -(R + B^T P_{\infty} B)^{-1} (B^T P_{\infty} A), \quad (9)$$

where P_{∞} is the unique positive definite solution of the Algebraic Riccati Equation (ARE)

$$P_{\infty} = A^T P_{\infty} A - (A^T P_{\infty} B) (R + B^T P_{\infty} B)^{-1} (B^T P_{\infty} A) + Q. \quad (10)$$

In our case, we have that $Q = 1$ and $R = \frac{3}{4}$. Since $A = 1$ and $B = 1$, it follows

$$P_{\infty} = P_{\infty} - P_{\infty} \left(\frac{3}{4} + P_{\infty} \right)^{-1} P_{\infty} + 1, \quad (11)$$

$$\Leftrightarrow 0 = P_{\infty}^2 - P_{\infty} - \frac{3}{4}, \quad (12)$$

where P_∞ is a scalar (generally, it is a matrix). This is a second order polynomial, which has two solutions

$$P_\infty = \frac{3}{2} \text{ or } P_\infty = -\frac{1}{2}. \quad (13)$$

The negative P_∞ has to be discarded since it is not positive definite. Note that there are no stability guarantees for any other solution apart from the unique positive definite solution. Finally, we substitute the P_∞ we found in the Eq.(9) and we get

$$F_\infty = -\frac{2}{3}. \quad (14)$$

Note: In this exercise, we computed P_∞ by hand. The ARE can be solved efficiently with the MATLAB command `d1qr`.

3. The resulting closed-loop linear system can be written as

$$x(k+1) = Ax(k) + BF_\infty x(k). \quad (15)$$

For a linear system, the LQR controller ensures asymptotic stability of the closed-loop system under the stabilizability and detectability assumptions (lecture 2, slide 42). We can verify that (A, B) is stabilizable and $(Q^{1/2}, A)$ is detectable with the definitions:

- Stabilizable (Lecture 1c, Slide 41): if $\text{rank}(\lambda_j I - A|B) = n$, where λ_j is any eigenvalue of A lying on or outside the unit circle (discrete-time) and n is the dimensionality of our state. Indeed, we have that $\text{rank}([1 - 1|1]) = \text{rank}([0|1]) = 1$.
- Detectable (Lecture 1c, Slide 44): if $\text{rank}(A^T - \lambda_j I|Q^{T/2}) = n$, where λ_j is any eigenvalue of A lying on or outside the unit circle (discrete-time) and n is the dimensionality of our state. Again, we have that $\text{rank}([1 - 1|1]) = \text{rank}([0|1]) = 1$.

Without using the explicitly computed LQR controller, we can conclude that the origin is asymptotically stable for the closed-loop linear system. We can also check this by explicitly computing the closed-loop system dynamics

$$x(k+1) = (A + BF_\infty)x(k) = \frac{1}{3}x(k). \quad (16)$$

Note: Since we obtained the LQR controller for the linearized system and the linearization is only valid close to the linearization point, the stability property of the closed-loop linear system concluded here holds only locally around the linearization point for the closed-loop nonlinear system.

Note: We can also show controllability and observability, which implies stabilizability and detectability.

4. The closed-loop nonlinear system can be written as

$$x(k+1) = x(k) + x(k)^2 + F_\infty x(k) \quad (17)$$

$$= \frac{1}{3}x(k) + x(k)^2. \quad (18)$$

For Lyapunov's direct method we need to find a Lyapunov function $V(x)$ which satisfies the following conditions (Slide 49 Lec.1c):

- $\|x\| \rightarrow \infty \Rightarrow V(x) \rightarrow \infty$.
- $V(0) = 0$ and $V(x) > 0, \forall x \in \mathbb{R} \setminus \{0\}$.
- $V(f(x)) - V(x) \leq -\alpha(x), \forall x \in \mathbb{R}$ where $\alpha(x)$ has to be continuous positive definite.

We assume that a suitable Lyapunov function may be $V(x) = x^2$. The first two conditions are fulfilled trivially and for the third one we calculate

$$V(x(k+1)) - V(x(k)) = \left(\frac{1}{3}x + x^2\right)^2 - x^2 \quad (19)$$

$$= -x^2\left(\frac{8}{9} - \frac{2}{3}x - x^2\right). \quad (20)$$

We want $V(x(k+1)) - V(x(k)) < 0, \forall x \neq 0$ and since it holds that $-x^2 < 0, \forall x \neq 0$, we need $\frac{8}{9} - \frac{2}{3}x - x^2 > 0, \forall x \neq 0$. Unfortunately, this is not possible because

$$\frac{8}{9} - \frac{2}{3}x - x^2 > 0 \Leftrightarrow x \in \underbrace{\left(-\frac{4}{3}, \frac{2}{3}\right)}_{\neq \mathbb{R} \setminus \{0\}}. \quad (21)$$

With our choice $V(x) = x^2$ we cannot conclude that the origin is globally asymptotically stable. However, this does not mean that there cannot be a different choice of $V(x)$ which is a valid Lyapunov function and proves global asympt. stability.